AI Financial Risk & Portfolio Intelligence Platform

S&P Global & CRISIL Campus Hackathon 2026

- **Candidate:** Basant Bansal
- **Campus:** Indian Institute of Information Technology, Pune (IIIT Pune)
- **Public Repository:** https://github.com/basantbansal/iiitpune-basant-bansal-hackathon
- **Architecture:** Hybrid AI/NLP Ingestion + Tactical Index Rebalancing + Wholesale Banking Book Stress Testing

1. Problem Statement & Solution Overview

The Challenge

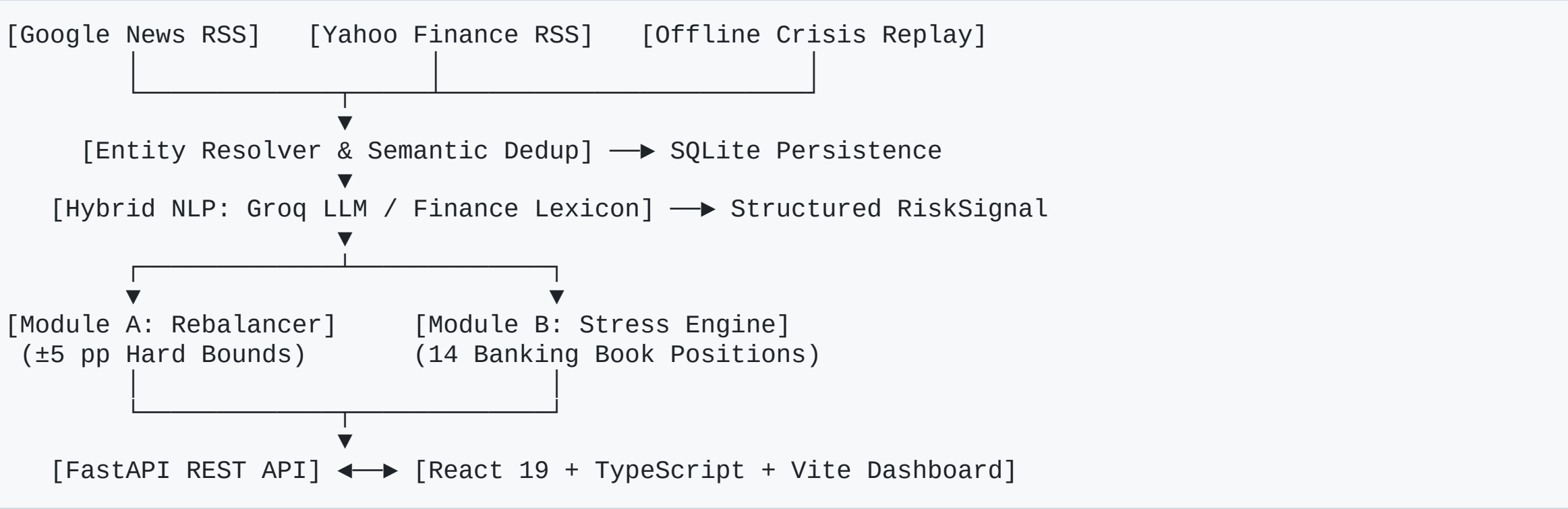
Financial markets absorb unstructured textual events (earnings releases, geopolitical tensions, credit shocks) before pricing models adjust. Traditional equal-weight or market-cap indices react with significant lag, amplifying drawdowns.

Our Solution

An end-to-end institutional financial risk intelligence platform:

1. **Dual Live Ingestion**: Live Google News RSS & Yahoo Finance RSS feeds with semantic deduplication.
2. **Explainable AI/NLP Risk Engine**: Groq LLaMA 3.3 70B (JSON mode) + deterministic financial lexicon fallback producing structured `RiskSignal` records.
3. **Module A (Tactical Rebalancer)**: Mean sentiment tilting bounded strictly to $10\% \pm 5\%$ ($[5\%, 15\%]$) with exact trade & fee accounting.
4. **Module B (Stress Engine)**: Multi-factor stress testing across equities & a 14-position Wholesale Banking Book with auto-triggering on high-impact signals.

2. System Architecture & Dual Ingestion



- **Dual Independent Feeds**: Google News RSS + Yahoo Finance RSS without proprietary vendor lock-in.
- **Enterprise Deduplication**: Cryptographic SHA-256 + 24-hour temporal deduplication window.

3. Structured NLP Risk Engine

Hybrid Architecture for High Reliability & Zero Hallucination

- **Primary LLM**: Groq API (`llama-3.3-70b-versatile`) with strict JSON schema and prompt grounding.
- **Deterministic Lexicon Scorer**: Finance-aware dictionary with negation detection, contrastive handling ("despite", "but"), and systemic risk weighting.
- **Abstention Policy**: Ambiguous or single-word matches output `Unclassified / Needs Review` to avoid false triggers.

Structured Output Schema (`RiskSignal`)

Field	Domain / Range	Example Output	Purpose
`entity`	Monitored ticker	`AAPL`	Constituent entity assignment
`sentiment_score`	Float $[-1.0, +1.0]$	`-0.65`	Tactical weight tilting factor
`event_classification`	8 Taxonomy categories	`Supply Chain / Operations`	Domain classification
`impact_score`	Integer $[1, 10]$	`8`	Tail risk & trigger prioritization
`confidence`	Float $[0.0, 1.0]$	`0.85`	Signal certainty filter
`evidence_span`	Snippet from headline	`"cuts component orders"`	Auditable attribution

4. Module A: Tactical Index Rebalancer

Algorithmic Rebalancing with Simplex Constraints

- **Universe**: S&P Top 10 constituents (`AAPL`, `MSFT`, `NVDA`, `AMZN`, `GOOGL`, `META`, `TSLA`, `JPM`, `JNJ`, `XOM`) with 10.0% baseline weight.
- **Strict Bounds**: Tracking error bounded to $\text{Baseline} \pm 5\%$ ($[5\%, 15\%]$); simplex constraint $\sum w_i = 1.0$.
- **Neutrality Invariant**: Constituents with no new usable signals strictly preserve baseline weights (absence of news $\neq$ negative sentiment).

Exact Accounting & Persistent Trajectory

- **Order Reconciliation**:

$$\text{Post-Trade Shares} = \text{Pre-Trade Shares} + \text{Signed Trade Quantity}$$

$$\text{Simulated Fees} = \text{Gross Traded Value} \times 10 \text{ bps } (\text{half-up rounded to } \$0.01)$$

- **Multi-Scan History**: SQLite persistence (`target_weight_history`) tracking weight trajectories over successive rebalance scans.

5. Module B: Macroeconomic & Credit Stress Testing

Multi-Asset Wholesale Banking Book (14 Positions)

- **Corporate Loans (4)**: Expected Loss $\Delta \text{EL} = -\text{Notional} \cdot \Delta \text{PD} \cdot \text{LGD}$ plus Credit Spread term.
- **Fixed Income Bonds (4)**: Price Change $= -\text{ModDur} \cdot \Delta \text{Rate} - \text{SpreadDur} \cdot \Delta \text{Spread}$.
- **Interest Rate Swaps (3)**: P&L $= \text{Signed DV01} \cdot \Delta \text{Rate bps}$ (Payer vs Receiver hedges).
- **Equity Holdings (3)**: Factor shock $= \text{Notional} \cdot (\beta \cdot \Delta \text{Equity})$.

Guaranteed Reconciliation & Auto-Trigger Logic

- **Exact Invariant**: $\text{Post-Shock Value} \equiv \text{Pre-Shock Value} + \sum \text{Position P\&L}$ across all asset classes.
- **Auto-Trigger**: Any signal with $\text{Impact} \geq 7$ auto-recommends a targeted stress scenario:
  - *Macroeconomic* $\rightarrow$ 2022 Fed Tightening Shock
  - *Credit / Geopolitical* $\rightarrow$ Systemic Credit Crunch
  - *Tech / Product* $\rightarrow$ Tech Bubble Burst

6. Validation, Data Provenance & Domain Impact

Engineering Rigor & Test Suite

- **Full Test Suite Passing**: Automated pytest verification across risk engine, order accounting, fee rounding, and banking book math.
- **Production Bundle**: Clean TypeScript compilation (`tsc -b`) and Vite production bundle.
- **Zero Secret Leaks**: Verified git history with no API keys or credentials committed.

Data Provenance Disclosures

- **Live Market News**: Real-time Google News RSS & Yahoo Finance RSS feeds.
- **Synthetic Banking Book**: 14 institutional positions (`data/synthetic_banking_book.json`).
- **Research Benchmark (`data/all-data.csv`)**: Financial PhraseBank dataset (4,846 rows, Latin-1); not required at runtime (runtime uses live feeds + Groq / deterministic rule engine).
- **Domain Impact**: Bridges qualitative news intelligence with quantitative risk management, providing institutional risk officers with auditable, real-time portfolio resilience.